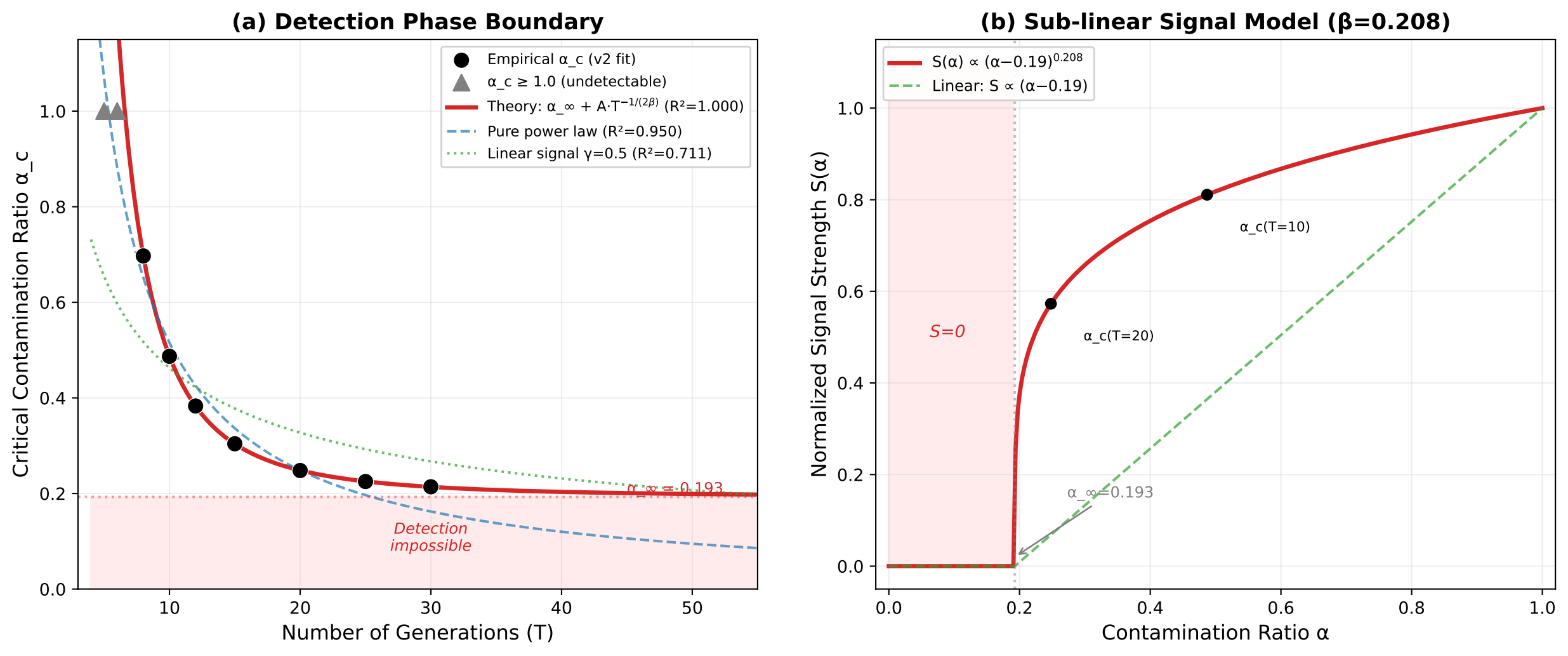
(a) Detection Phase Boundary
Empirical α_c (v2 fit)
α_c ≥ 1.0 (undetectable)
Theory: α_∞ + A·T^{-1/(2β)} (R²=1.000)
Pure power law (R²=0.950)
Linear signal γ=0.5 (R²=0.711)
α_∞ = 0.193
Detection impossible
Critical Contamination Ratio α_c
Number of Generations (T)
0.0
0.2
0.4
0.6
0.8
1.0
10
20
30
40
50
(b) Sub-linear Signal Model (β=0.208)
S(α) ∝ (α−0.19)^{0.208}
Linear: S ∝ (α−0.19)
S=0
α_c(T=10)
α_c(T=20)
α_∞=0.193
Normalized Signal Strength S(α)
Contamination Ratio α
0.0
0.2
0.4
0.6
0.8
1.0